

Module 1 – Introduction to Machine Learning

1. What are the issues in Machine learning?
 2. Explain the steps of developing Machine Learning applications.
 3. How to choose the right ML algorithm?
 4. Explain any five applications of Machine Learning.
 5. Explain Training error and Generalization error.
 6. Differentiate between Supervised and Unsupervised Learning.
 7. Demonstrate steps to design a Machine Learning application.
 8. What is over fitting, under fitting and Bias-variance trade-off with reference to Machine learning?
 9. Explain the terms overfitting, underfitting, bias & variance trade-off w.r.t. Machine Learning.
 10. Explain any five business applications of Machine learning.
 11. Write a short note on issues in Machine Learning.
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Module 2 – Learning with Regression and Trees

1. Explain Regression line, Scatter plot, Error in prediction and Best fitting line.
2. Explain Logistic Regression.
3. Explain Linear regression along with an example.
4. Create a decision tree using Gini Index to classify following dataset:

Sr. No.	Income	Age	Own
1	Very High	Young	Yes
2	High	Medium	Yes
3	Low	Young	No
4	High	Medium	Yes
5	Very High	Medium	Yes
6	Medium	Young	Yes
7	High	Old	No
8	Medium	Medium	No
9	Low	Medium	Yes
10	Low	Old	No
11	High	Young	No
12	Medium	Old	Yes

5. Explain Multivariate Linear regression method.
6. Create a decision tree using Gini Index to classify dataset for profit.

Age	Competition	Type	Profit
old	software	down	Yes
old	software	Down	No
old	hardware	Down	No
mid	software	Down	Yes
mid	hardware	Down	Yes
mid	hardware	Up	No
mid	software	Up	No
new	software	Up	Yes
new	hardware	Up	No
new	software	Up	No

7. Demonstrate CART method along with example.

8. Following table shows midterm and final exam grades; use linear regression to predict:

Midterm (X)	72	50	81	74	94	86	59	83	86
Final (Y)	84	53	77	78	90	75	49	79	77

9. Differentiate between Linear regression and Logistic regression.

10. Consider example where mass y (grams) relates to time x (seconds): find regression line.

Time (x)	5	7	12	16	20
Mass (y)	40	120	180	210	240

11. Explain concept of decision tree; compute Gini Index for features.

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Negative	High	Low	Down
Positive	High	High	Up
Negative	Low	High	Down
Positive	Low	Low	Down
Positive	High	High	Up

12. Create a decision tree using Gini Index for “car stolen?” dataset.

Car no	Colour	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes

Car no	Colour	Type	Origin	Stolen?
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

13. Explain Gini index along with an example.
14. Write detailed note on Linear Regression.
15. Explain Performance Metrics for Classification.

Module 3 – Ensemble Learning

1. Explain the Random Forest algorithm in detail.
2. Explain the different ways to combine the classifiers.
3. Explain the concept of bagging and boosting.
4. Compare Bagging and Boosting with reference to ensemble learning.
5. Explain Random Forest and its use cases.
6. Demonstrate Ensemble learning-based Random Forest algorithm in detail.
7. Write a short note on XGBoost ensemble method.
8. Explain necessity of cross validation in Machine Learning applications and K-fold cross validation in detail.

Module 4 – Learning with Classification

1. Explain the concept of margin and support vector.
2. Describe Multiclass classification.
3. Explain support vector machine as constraint optimization problem.
4. Explain SVM Kernel trick.
5. Define Hyper plane, Support Vectors, Hard Margin, Soft Margin, Kernel.
6. Differentiate between Logistic regression and Support vector machine.
7. Explain SVM as a constrained optimization problem.
8. Explain Kernel Trick in SVM.
9. Explain multiclass classification techniques.
10. Explain performance evaluation metrics for binary classification with example.
11. Explain Accuracy, Precision, Recall and F1 Score with example.
12. Consider the use case of Email spam detection. Identify and explain the suitable machine learning technique for this task.

Module 5 – Learning with Clustering

1. Explain the distance metrics used in clustering.
2. Explain clustering with minimal spanning tree with reference to Graph based clustering.
3. Demonstrate MST algorithm along with example.
4. Explain Clustering with minimal spanning tree along with example.
5. Explain the concept of Expectation Maximization Algorithm.
6. Explain EM algorithm.
7. Explain DBSCAN algorithm along with example.
8. Demonstrate DBSCAN algorithm along with example.
9. What is Density-based clustering? Explain DBSCAN.

Color	Wig	Num.Ears	Emotion
G	Y	2	S
G	N	2	S
B	N	2	H
R	Y	3	H

Module 6 – Dimensionality Reduction

1. Principal Component Analysis for Dimension Reduction.
2. Compute Linear Discriminant projection for given two-dimensional dataset:
 - $X_1 = \{(4, 1), (2, 4), (2, 3), (3, 6), (4, 4)\}$
 - $X_2 = \{(9, 10), (6, 8), (9, 5), (8, 7), (10, 8)\}$
3. Write short note on Linear Discriminant projection along with example.
4. Give steps to design PCA dimensional reduction technique with example.
5. Explain Principal Component Analysis for Dimensionality Reduction in detail.
6. Explain Linear Discriminant Analysis and its real-world applications.
7. Explain Linear Discriminant Analysis.
8. Explain the following: Receiver Operating Characteristics curve and Area Under Curve.
9. What is dimensionality reduction? Explain how it can be utilized for classification and clustering tasks.